ITMA - 0891 - NAJ - Junior.

Answer All Questions.

1. The value of $(24.681)^2 - (14.681)^2$ is

a) 39.362 b) 3.9362 c) 393.62 d) 3936.2

2. 121×1001 is equal to

a) 122121 b) 121121 c) 112211 d) 121221

3. 0.4366.... is the value of the fraction

a) $\frac{14}{30}$ b) $\frac{141}{300}$ c) $\frac{131}{300}$ d) $\frac{132}{300}$

4. If $\log_3 5 = x$, then $4x + 3$ is the value of

a) $\log_3 16785$ b) $\log_3 16875$ c) $\log_3 17865$ d) $\log_3 16865$

5. If $(x-2), (x-3)$ are the factors of the equation $2x^3 - 9x^2 + ax + b$ then a,b are respectively

a) $-7, -6$ b) $-7, 6$ c) $7, -6$ d) $7, 6$

6. The sum of the squares of two consecutive natural numbers is 19405. The numbers are

a) 98 , 99 b) 101 , 102 c) 103 , 104 d) 96 , 97

7. The sum of the first n odd natural numbers is

a) $\frac{n(n-1)}{2}$ b) $\frac{n(2n-1)}{2}$ c) $2n^2$ d) n^2

8. If $f(x) = \log_{10}(\frac{1+x}{1-x})$, then $f(\frac{2x}{1+x})$ is.

a) $(f(x))^2$ b) $f(2x)$ c) $2f(x)$ d) $\frac{f(x)}{2}$

9. A rectangular plot R_1 at one corner of another rectangular plot R_2 has 4 m.length and 3 m.breadth. If the diagonal of R_1 lies along the diagonal of R_2 and the breadth of R_2 is 15 m. the area of R_2 is

a) 180 sq.m. b) 240 sq.m. c) 300 sq.m. d) 320 sq.m.

10. Let C be a circle of radius r units. From a point A at a distance d units from the centre, a tangent is drawn to the circle. The length of the tangent is

a) $(d^2 + r^2)^{1/2}$ b) $(d^2 + r^2)$ c) $(d^2 - r^2)^{1/2}$ d) $(d^2 - r^2)$

11. The number of digits in $3^2 \times 2^8$ is

a) 9 b) 10 c) 8 d) 12

12. If the radius and height of a cylinder are equal to that of a cone, and the volume of the cone is 100 cm^3, then the volume of the cylinder is

a) $33\frac{1}{3}$ cm^3 b) $66\frac{2}{3}$ cm^3 c) 100 cm^3 d) 300 cm^3

13. If $3x + 4y = 13xy = 5x - 2y$, the value of x,y respectively are

a) $\frac{1}{3}, 1$ b) $1, \frac{1}{3}$ c) 3,1 d) 1,3

14. A certain fraction becomes 6 if 2 is subtracted from the numerator; it becomes 2 if 3 is added to the denominator. The fraction is

a) $\frac{16}{7}$ b) $\frac{20}{7}$ c) 8 d) $\frac{5}{8}$

15. A father is 7 times older than his son now. In 10 years, he will be 3 times older than his son. The present age of the father is

a) 28 b) 35 c) 42 d) 49

16. The difference between the simple interest and compound interest on a sum of Rs.5250 for 2 years at 6% per annum is

a) Rs.18.90 b) 19.80 c) Rs.189 d) Rs.198

17. In a triangle ABC, XY cuts AB, AC at X,Y respectively. Which of the following sets of lengths make XY parallel to BC?

a) AB = 12, XB = 4, AC = 8, AY = 6

b) AX = 6, XB = 5, AY = 9, YC = 8

c) AC = 21, YC = 9, AB = 14, AX = 8

d) AY = 6, AC = 13, XB = 5, AX = 4

18. A set A has 7 elements, B has 5 elements and $A \cap B$ has 2 elements. The number of elements in the set $A \cup B$ is

a) 12 b) 10 c) 14 d) cannot be determined by the given data.

19. The surface area of a cube is 150 sq.cms. The volume of the cube in cubic cms is

a) 25 b) 20 c) 5 d) 15

20. The diameters of the earth and the moon are in the ratio 4:1, approximtely. Their volumes are in the ratio

a) 4:1 b) 1:16 c) 64:1 d) 1:4

21. x,y are natural numbers such that $x^2y^3 = 72$; which of thefollowing statements is true?

a) x = 2 b) x = 6 c) The value cannot be determined as there are two unknowns but only one equation.
d) x = 3

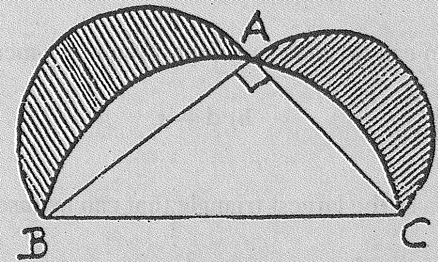
22. If $\sin\theta = 3/5$, the value of $\frac{2\tan\Theta}{1-\tan\Theta}$ is given by

a) $\frac{24}{25}$ b) $\frac{24}{7}$ c) $\frac{25}{24}$ d) $\frac{7}{24}$

23. A circle S is inscribed in a square ABCD of side r and a square LMNP is inscribed in S. The area of LMNP is

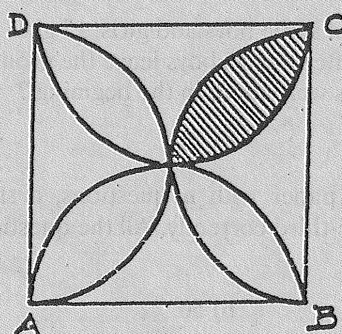
a) $\frac{r}{2}$ b) $\frac{r}{2}$ c) $\frac{r}{2}$ d) $\frac{r}{2}$

24. ABC is a triangle right angled at A and semi-circles are drawn on sides AB and AC asdiameters. (see figure) The area of the shaded region is that of



a) semi-circle ABC
b) triangle ABC
c) semi-circle on AC as diameter
d) semi-circle on AB as diameter.

25. ABCD is a square of side r and semi-circles are drawn on the sides as in the figure. The area of the shaded region is



a) $\frac{r}{2}(\pi-2)$ b) $\frac{r}{4}(\pi-2)$

c) $\frac{r}{8}(\pi-2)$ d) None of these.

26. m and n are positive integers such that $(\sqrt{3}+1)^m(\sqrt{3}-1)^n$ is a rational number. Then

a) m = n b) m < n c) m > n d) Nothing can be said about the relative sizes of m and n.

27. In the following m and n are natural numbers:

List 1	List 2
A) If n divides both m and m + 1, then	1) n = m
B) If n divides both m and m + 2, then	2) n = 1 or 3
C) If n divides both m and m + 3, then	3) n = 1 or 2
	4) n = 1

Now choose the correct matching from the following:

a)	b)	c)	d)
A – 1	A – 2	A – 4	A – 1
B – 2	B – 3	B – 3	B – 3
C – 3	C – 4	C – 2	C – 4

28. ABC is a triangle with sides AB = 6 cm, BC = 8 cm and CA = 10 cm; Then the area of the triangle in sq.cm. is

a) 24 b) 40 c) 30 d) cannot be found

29. AD, BE, CF are the altitudes of a triangle ABC such that AD = BE < CF. Which of the following is correct?

a) BC = CA > AB; b) BC = AC < AB c) BC > AC = AB d) BC < AC = BA

30. If the radius of a circle of radius A is doubled, then the area is increased by

a) A b) 2A c) 3A d) 4A

31. If m men can do a job in d days, then m + r men can do it in how many days ?

a) d + r b) d − r c) $\frac{md}{m+r}$ d) $\frac{d}{m+r}$

32. The area of the largest triangle that can be inscribed in a semi circel of radius r is

a) r^2 b) $2r^2$ c) $2r^3$ d) $\frac{1}{2}r^2$

33. In how many ways can a rectangle be divided into two equal parts ?

a) 2 b) 4 c) 32 d) uncountable.

34. From a group of some boys and girls, 15 girls leave first. Then the ratio of the number of girls to the number of boys become 1:2. After this 45 boys leave the group. Then the ratio of the number of boys to the number of girls becomes 1:5 How many girls were there in the begining ?

a) 40 b) 45 c) 30 d) 100

35. In a question paper with n questions, a student answers correctly 15 of the first 20. Of the remaining questions he answers one-third correctly. All the questions carry equal marks. If the student's mark is 50%, then what is the value of n?

a) 60 b) 50 c) 40 d) 100

36. Two boys A and B start at the same time to ride on bicycles with uniform speed from place P_1 to the place P_2, 60 km away. A travels 4 km an hour slower than B. B reaches P_2 and at once turns back meeting A, 12 km from P_2. Find the speed of A in km/hr.

a) 16 b) 8 c) 24 d) 60

37. The number of terms in the expansion of $[(x-3y)^2 (x+3y)^2]^2$ when simplified is

a) 4 b) 5 c) 7 d) more than 7

38. The botom, side and front areas of a rectangular box are known. The product of these areas is equal to

a) the volume of the box

b) the square root of the volume

c) twice the volume

d) the square of the volume

39. The base of an isosceles triangel is $\sqrt{2}$; the medians to legs intersect each other at right angles. The area of the triangle is

a) 1.5 b) 2 c) 2.5 d) 3.5

40. The angle of elevation of the top of a tower from a place is 45°. If the tower is 25m in height. how far is the place from the tower ?

a) $25\sqrt{2}$ m; b) $\frac{25}{2}$m c) 25 m d) 50 m.